Sequences & Patterns

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Find the 10th term of the sequence $5, 8, 11, 14, \ldots$

$a = 5$, $d = 3$. $a_{10} = 5 + 9 \times 3 = 32$.

2 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.

$a_{20} = 4 + 19 \times 6 = 4 + 114 = 118$.

3 [Foundational] • Target: Arithmetic sequences

The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.

$3 + 4d = 23 \Rightarrow 4d = 20 \Rightarrow d = 5$.

4 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?

$7 + (n - 1) \times 5 = 82 \Rightarrow n - 1 = 15 \Rightarrow n = 16$. The 16th term.

5 [Foundational] • Target: Arithmetic sequences

The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.

$a + 7 \times 4 = 34 \Rightarrow a + 28 = 34 \Rightarrow a = 6$.

6 [Foundational] • Target: Geometric sequences

Find the next term of the sequence $2, 4, 8, 16, \ldots$

Geometric, ratio 2: $16 \times 2 = 32$.

7 [Foundational] • Target: Geometric sequences

A geometric sequence has first term 5 and common ratio 3. Find the 5th term.

$a_5 = 5 \times 3^4 = 5 \times 81 = 405$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Geometric sequences

The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.

$2 \times r^3 = 54 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

9 [Intermediate] • Target: Geometric sequences

A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.

$r = 18 \div 6 = 3$. First term $= 6 \div 3 = 2$.

10 [Intermediate] • Target: Identifying the type of sequence

Is $5, 10, 20, 40, \ldots$ arithmetic, geometric, or neither?

Each term is double the previous one ($r = 2$): geometric.

11 [Intermediate] • Target: Finding missing terms

Find the missing term: $3,\ __,\ 11,\ 15$ (arithmetic).

Common difference $= (15 - 3) \div 3 = 4$. Missing term $= 3 + 4 = 7$.

12 [Intermediate] • Target: Finding missing terms

Find the missing term: $4,\ __,\ 36$ (geometric, all terms positive).

$r = \sqrt{36 \div 4} = \sqrt{9} = 3$. Missing term $= 4 \times 3 = 12$.

13 [Intermediate] • Target: Finding missing terms

Insert two arithmetic means between 5 and 20, forming the four-term sequence $5, __, __, 20$.

Common difference $= (20 - 5) \div 3 = 5$. The sequence is $5, 10, 15, 20$: the missing terms are 10 and 15.

14 [Intermediate] • Target: Finding missing terms

The sequence $x,\ 2x + 1,\ 4x - 1$ is arithmetic. Find x.

Equal consecutive differences: $(2x + 1) - x = (4x - 1) - (2x + 1) \Rightarrow x + 1 = 2x - 2 \Rightarrow x = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding missing terms • Cross-Lesson: 7.01 Data Analysis & Probability

Find the two missing terms of the geometric sequence $2,\ __,\ __,\ 54$.

$r^3 = 54 \div 2 = 27 \Rightarrow r = 3$. The sequence is $2, 6, 18, 54$: the missing terms are 6 and 18.

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16 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the sum of the first 10 terms of the sequence $2, 5, 8, 11, \ldots$

$a = 2$, $d = 3$, $n = 10$. $S = \frac{n}{2}(2a + (n-1)d) = 5(4 + 27) = 5 \times 31 = 155$.

17 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find the sum of the first 20 positive integers.

$S = \frac{20 \times 21}{2} = 210$.

18 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.

$S = \frac{15}{2}(2 \times 3 + 14 \times 4) = 7.5 \times (6 + 56) = 7.5 \times 62 = 465$.

19 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 3.01 Algebra Essentials

Find the sum of the first 25 positive even integers $(2, 4, \ldots, 50)$.

$a = 2$, $d = 2$, $n = 25$. $S = \frac{25}{2}(4 + 24 \times 2) = 12.5 \times 52 = 650$.

20 [Challenging] • Target: Special sequences • Cross-Lesson: 4.01 Geometry & Measurement

The sequence $1, 4, 9, 16, 25, \ldots$ consists of perfect squares. Find the 8th term.

The nth term is n^2, so the 8th term is $8^2 = 64$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Visual and growing patterns

The bar chart shows the number of tiles used to build patterns 1 through 4. Predict the number of tiles in pattern 8.

a. 37

b. 40

c. 29

d. 33 ✓

$d = 4$, $a_1 = 5$: $T_8 = 5 + 7(4) = 33$.

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22 [Foundational] • Target: Visual and growing patterns

The bar chart shows the number of matchsticks used to build squares in a row, for 1 through 4 squares. Find the number needed for 12 squares.

a. 34

b. 43

c. 40

d. 37 ✓

$d = 3$, $a_1 = 4$: $T_{12} = 4 + 11(3) = 37$.

23 [Foundational] • Target: Visual and growing patterns

The bar chart shows the number of matchsticks used to build hexagons in a row, for 1 through 4 hexagons. Find the number needed for 9 hexagons.

a. 41

b. 51

c. 50

d. 46 ✓

$d = 5$, $a_1 = 6$: $T_9 = 6 + 8(5) = 46$.

24 [Foundational] • Target: Geometric sequences

A geometric sequence has 3rd term 20 and 4th term 40. Find the first term and the common ratio.

a. $a_1 = 5, r = 2$ ✓

b. $a_1 = 20, r = 2$

c. $a_1 = 5, r = 4$

d. $a_1 = 10, r = 2$

$r = 40 \div 20 = 2$; $a_1 = 20 \div r^2 = 20 \div 4 = 5$.

25 [Foundational] • Target: Finding missing terms

The sequence $2x,\ 3x + 4,\ 5x + 2$ is arithmetic. Find x.

a. 4

b. 2

c. 8

d. 6 ✓

$(3x + 4) - 2x = (5x + 2) - (3x + 4) \Rightarrow x + 4 = 2x - 2 \Rightarrow x = 6$.

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26 [Foundational] • Target: Finding missing terms

Find the missing term: $__,\ 12,\ 19,\ 26$ (arithmetic).

a. 7

b. 9

c. 4

d. 5 ✓

$d = 7$: missing $= 12 - 7 = 5$.

27 [Foundational] • Target: Special sequences

Find the 7th triangular number, from the sequence $1, 3, 6, 10, 15, 21, \ldots$

a. 28 ✓

b. 25

c. 36

d. 21

$T_7 = \frac{7(8)}{2} = 28$.

28 [Foundational] • Target: Special sequences

A sequence starts $3, 5, 8, 13, 21, \ldots$, where each term is the sum of the two terms before it. Find the 8th term.

a. 55

b. 89 ✓

c. 144

d. 76

Continuing: $3, 5, 8, 13, 21, 34, 55, 89$: the 8th term is 89.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Word problems

A theater has 18 seats in the first row, and each following row has 4 more seats than the row before. How many seats are in row 15?

a. 78

b. 70

c. 56

d. 74 ✓

$a_1 = 18$, $d = 4$: $T_{15} = 18 + 14(4) = 74$.

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30 [Intermediate] • Target: Word problems

A ball is dropped and after each bounce rises to half its previous height. If it first rises to 120 cm after the first bounce, how high does it rise after the 5th bounce?

a. 60

b. 15

c. 3.75

d. 7.5 ✓

$a_1 = 120$, $r = 0.5$: $T_5 = 120(0.5)^4 = 7.5$ cm.

31 [Intermediate] • Target: Word problems

A savings plan deposits 80 riyals in week 1, then increases the deposit by 15 riyals each following week. Find the deposit in week 10.

a. 215 ✓

b. 200

c. 230

d. 935

$a_1 = 80$, $d = 15$: $T_{10} = 80 + 9(15) = 215$ riyals.

32 [Intermediate] • Target: Word problems

A bacteria culture starts with 40 bacteria and triples every hour. How many bacteria are there after 4 hours?

a. 1,080

b. 3,240 ✓

c. 9,720

d. 480

$40 \times 3^4 = 40 \times 81 = 3{,}240$.

33 [Intermediate] • Target: Pattern reasoning

The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in a polygon with 9 sides.

a. 36

b. 27 ✓

c. 21

d. 54

$\frac{9(6)}{2} = 27$.

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34 [Intermediate] • Target: Pattern reasoning

A sequence follows the rule \"triple and subtract 2\", starting at 4 as the 1st term. Find the 4th term.

a. 34

b. 28

c. 82 ✓

d. 250

$4 \to 10 \to 28 \to 82$: the 4th term is 82.

35 [Intermediate] • Target: Pattern reasoning

The nth term of a sequence is $T_n = 4n - 3$. Find the 15th term, and state whether the sequence is arithmetic or geometric.

a. 57, geometric

b. 53, arithmetic

c. 60, arithmetic

d. 57**, arithmetic** ✓

$T_{15} = 4(15) - 3 = 57$; since T_n is linear in n, it's arithmetic.

36 [Intermediate] • Target: Pattern reasoning

The nth term of a sequence is $T_n = 5(2)^{n-1}$. Find the 6th term, and state whether the sequence is arithmetic or geometric.

a. 160**, geometric** ✓

b. 320, geometric

c. 160, arithmetic

d. 80, geometric

$T_6 = 5(2^5) = 160$; since T_n involves a constant raised to the nth power, it's geometric.

37 [Intermediate] • Target: Sequences & Patterns

In a laboratory calibration sequence, find the 13th term of the sequence $-14,\ -21,\ -28,\ -35,\ \ldots$

a. -98 ✓

b. -105

c. -96

d. -91

$a = -14$, $d = -7$. $a_{13} = -14 + 12 \times (-7) = -98$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

During an engineering benchmark, find the 20th term of the sequence $1, -3, -7, -11, \ldots$

a. -71

b. -75 ✓

c. -73

d. -79

$a = 1, d = -4$. $a_{20} = 1 + 19 \times (-4) = -75$.

39 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 3.01 Algebra Essentials

In a telemetry monitoring trial, find the 7th term of the sequence $-5, -1, 3, 7, \ldots$

a. 23

b. 19 ✓

c. 21

d. 15

$a = -5, d = 4$. $a_7 = -5 + 6 \times 4 = 19$.

40 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 4.01 Geometry & Measurement

While auditing system throughput, find the 6th term of the sequence $-2, 1, 4, 7, \ldots$

a. 10

b. 13 ✓

c. 16

d. 15

$a = -2, d = 3$. $a_6 = -2 + 5 \times 3 = 13$.

41 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 5.01 Quantitative Comparisons

Find the 3rd term of the sequence $-3, 6, -12, 24, \ldots$

a. 12

b. -10

c. -14

d. -12 ✓

$a = -3, r = -2$. $a_3 = -3 \times (-2)^2 = -3 \times 4 = -12$.

42 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 7.01 Data Analysis & Probability

Under steady-state operating conditions, find the 7th term of the sequence $-1, 3, -9, 27, \ldots$

a. -729 ✓

b. 729

c. -727

d. -731

$a = -1, r = -3$. $a_7 = -1 \times (-3)^6 = -1 \times 729 = -729$.

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43 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 8.01 Word Problems & Reasoning

Across successive production cycles, find the 5th term of the sequence $3, -9, 27, -81, \ldots$

a. -243

b. 245

c. 241

d. 243 ✓

$a = 3, r = -3$. $a_5 = 3 \times (-3)^4 = 3 \times 81 = 243$.

44 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 1.01 Arithmetic & Number Sense

In a precision metrology test, find the 3rd term of the sequence $7, 21, 63, 189, \ldots$

a. 61

b. 65

c. 63 ✓

d. -63

$a = 7, r = 3$. $a_3 = 7 \times 3^2 = 7 \times 9 = 63$.

45 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

For a control-loop diagnostic run, find the 3rd term of the sequence $-1, -3, -9, -27, \ldots$

a. 9

b. -9 ✓

c. -7

d. -11

$a = -1, r = 3$. $a_3 = -1 \times 3^2 = -1 \times 9 = -9$.

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